Introduction

What is population genetics? what does it study?

Population genetics

- A population is a group of individuals from the same species that lives in the same geographic area and that actually or potentially interbreeds.
- Population genetics studies the genetic variation in populations and how it changes over time

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Why study population genetics?

Biological evolution is the change over time in the **genetic composition of a population**, which changes due to the birth and death of individuals or the migration of individuals in or out of the population.

What are the population events driving biological evolution?

- birth
- death
- migration

What are the forces driving biological evolution?

- genetic drift
- natural selection
- migration
- ...

Types of genetic variation

- SNP

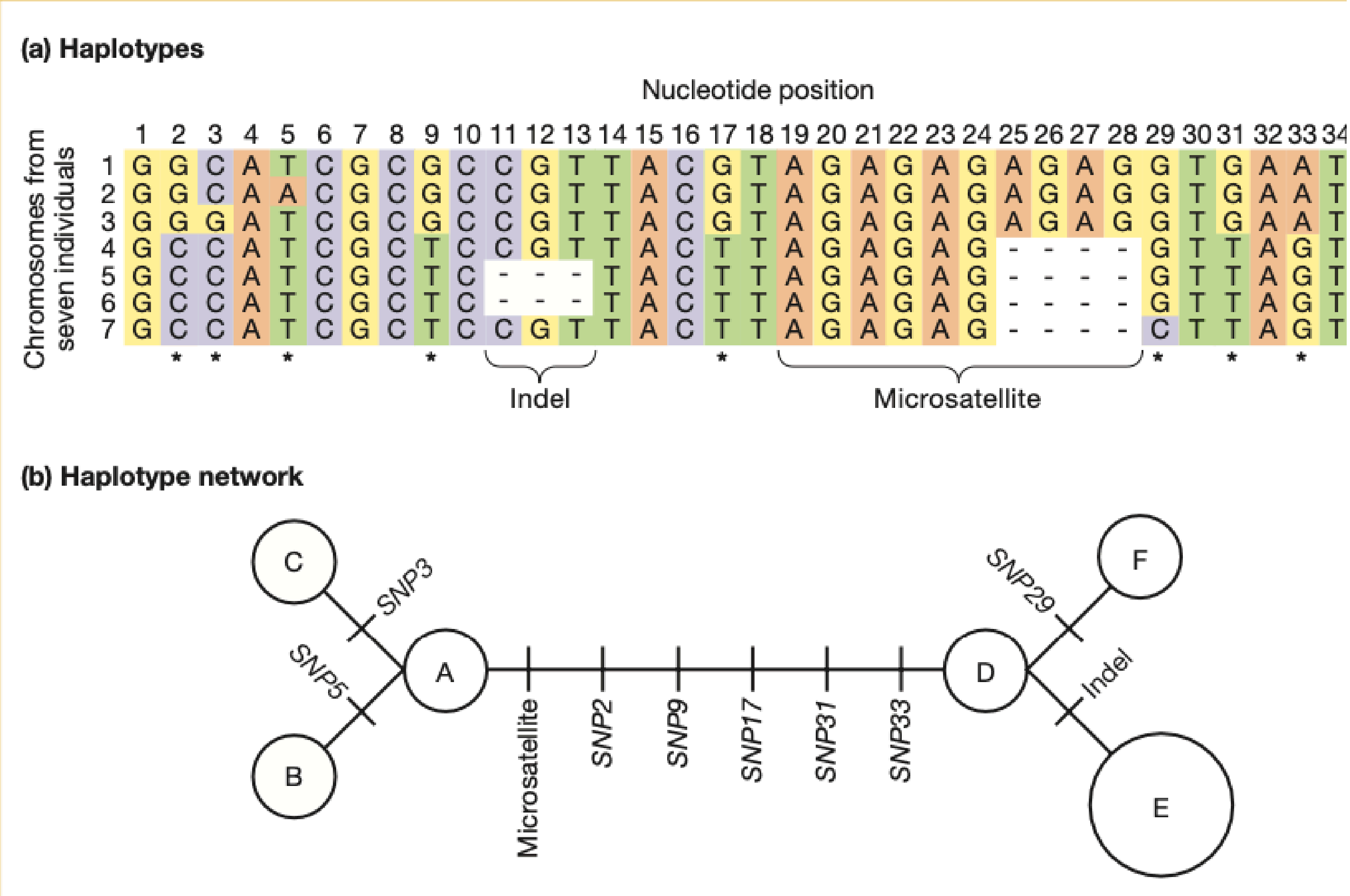
- microsatellites
    - What are microsatellites?
    - what are they useful for?
- CNV

Types of genetic variation

> **Haplotype**
>
> refer to the combination of alleles at multiple loci on the same chromosomal homolog.

How do you draw the network (b) based on the alignment (a)?

A haplotype network shows the relationship among haplotypes



<%? footnotes %>

[1]: Introduction to Genetic Analysis, ed 11, Fig. 18-4

note:

- comment on the different elements in the network map, e.g., circles, edges
- why do haplotypes exist? why do we have combinations of alleles?
- what breaks linkage?

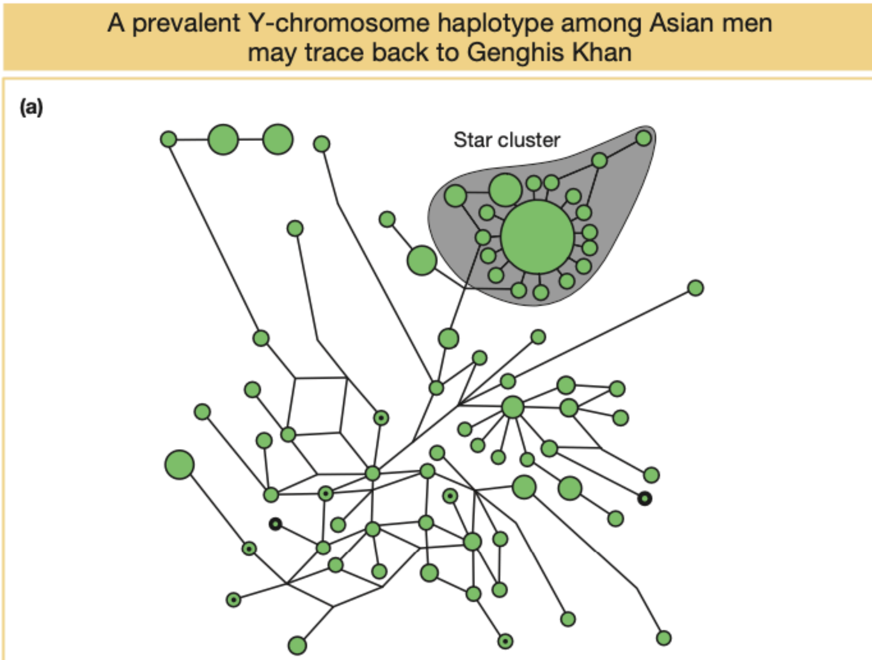


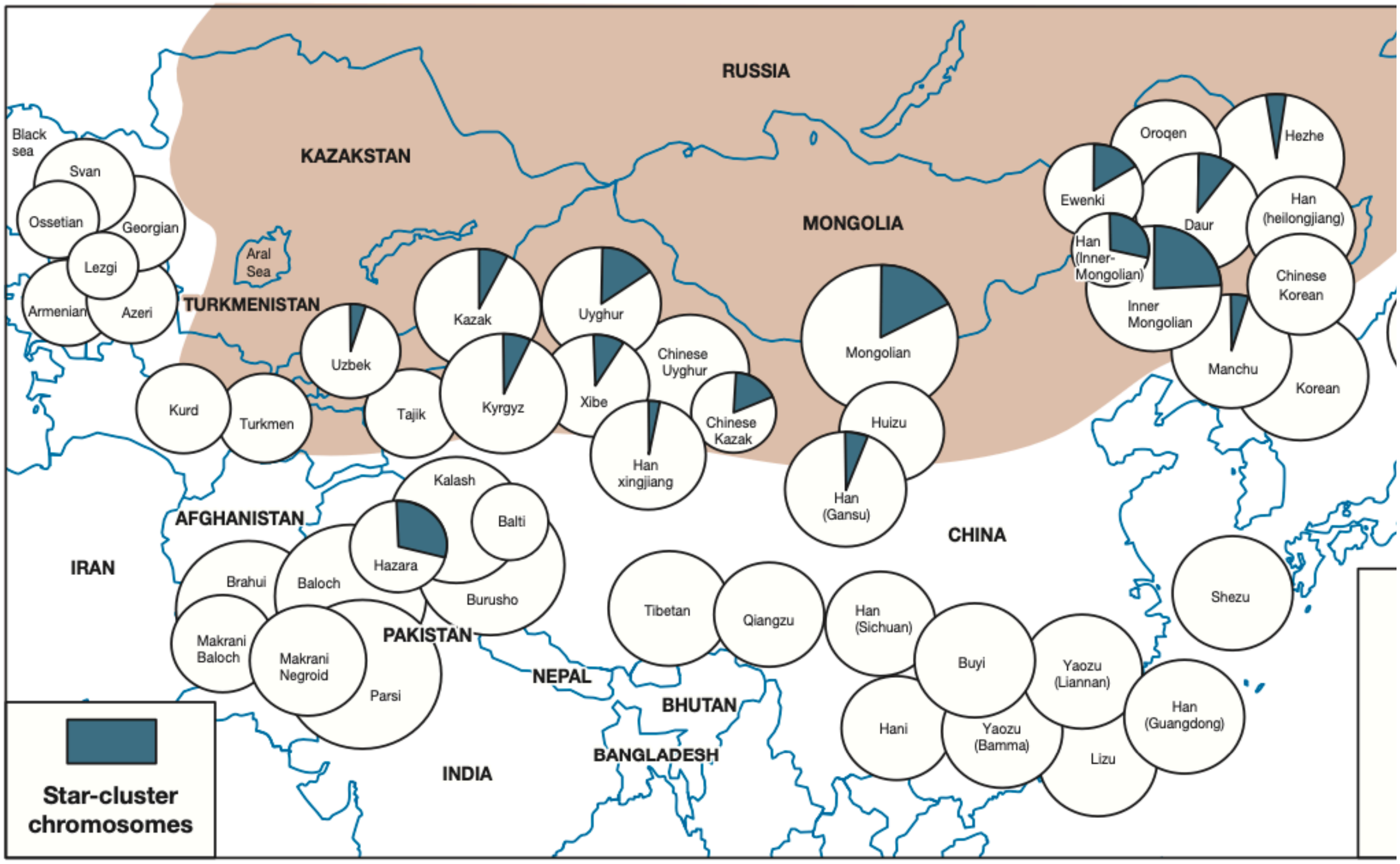
FIGURE 18-5 (a) Haplotype network for the Y chromosomes of Asian men showing the predominance of the star-cluster haplotype thought to trace back to Genghis Khan. The area of the circle is proportional to the number of individuals with the specific haplotype that the circle represents. (b) Geographical distribution of the star-cluster haplotype. Populations are shown as circles with an area proportional to sample size; the proportion of individuals in the sample carrying star-cluster chromosomes is indicated by green sectors. No star-cluster chromosomes were found in populations having no green sector in the circle. The shaded area represents the extent of Genghis Khan's empire. [*Data from T. Zerjal et al.,* Am. J. Hum. Genet. *72, 2003, 717–721.*]

<%? footnotes %>

note: Most men have a rare Y chromosome haplotype (why?)

Distribution of the star-cluster haplotype

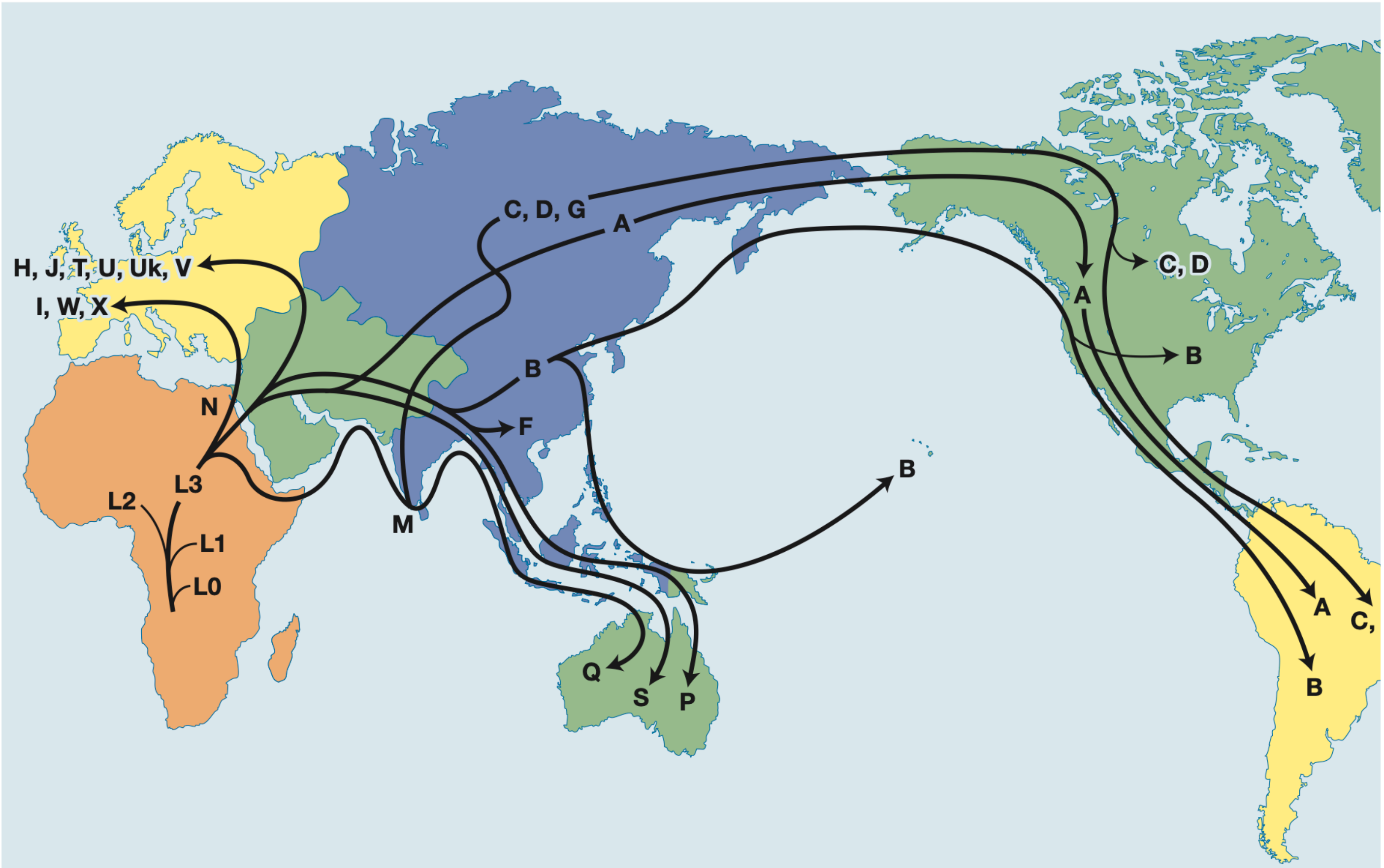
(b)



<%? footnotes %>

Where do we come from?

Mitochondrial haplotypes can be used to trace human origins to Africa



<%? footnotes %>

Gene pool

- We have been evaluating genetic variations from the individual's perspective ("horizontally" as a haplotype)
- We can also look from a site / locus / gene perspective ("vertically").
- Introduce the gene pool concept

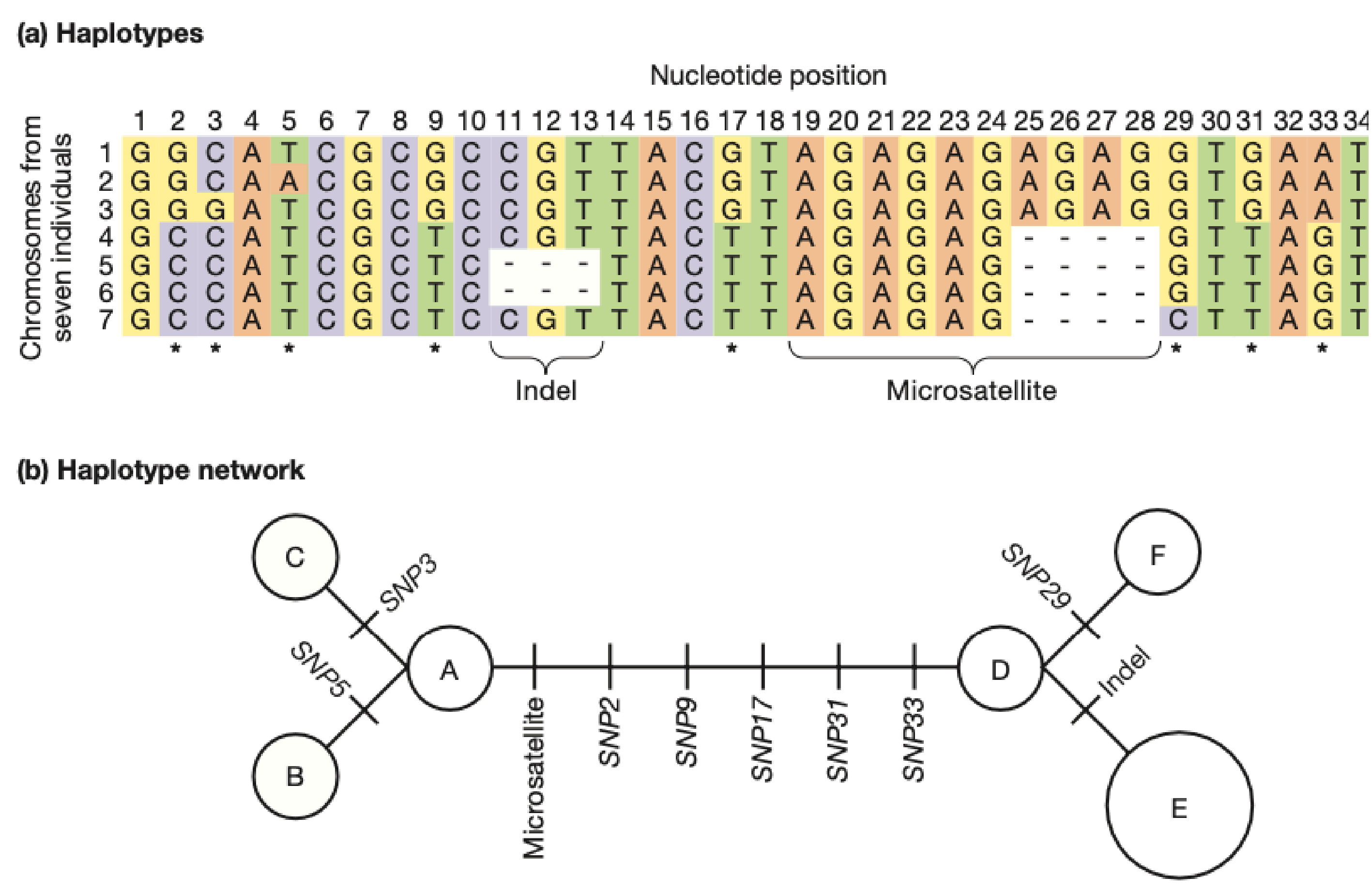
Gene pool

Gene pool

Sum total of all alleles in the breeding members of a population at a given time.

Gene pool and allele frequencies

A haplotype network shows the relationship among haplotypes



- **Locus**: "region of interest", may be an entire gene, or a single nucleotide
- **Allele**: different "versions" of the sequence existing in a population
- How many alleles are there if we take the entire haplotype region as a locus? what's the frequency of allele E (II-b)?
- How about just looking at site 9? What's the frequency of G?

Gene pool and genotype frequencies

- In diploid organisms, **genotype** denotes the total allele combinations at a locus (AA, Aa, aa).

- How do genotype frequencies depend on allele frequencies (p_A and p_a)?
- Why do we care?

- Genetic evolution = change in allele frequencies.
- Natural selection acts on fitness <- phenotype <- genotype

Gene pool and genotype frequencies

- **Locus**: "region of interest", may be an entire gene, or a single nucleotide
- **Allele**: different "versions" of the sequence existing in a population
- **Genotype**: denotes the total allele combinations at a locus, e.g., two alleles for diploids.

- What are the genotype frequencies?
- What are the allele frequencies?

Gene pool and genotype frequencies

- Knowing the allele frequency ($p_A = 18/32$), what is a "reasonable guess" of the genotype frequency (e.g., for $f_{A/A}$)?

- How close is it to the observed genotype frequency?
- Now reflect on your intuitive guess, what assumptions need to be made for it to work?

Hardy-Weinberg Law

Suppose allele frequency of $A = p$, and frequency of $a = q$

$f_{A/A}$	$f_{A/a}$	$f_{a/a}$
p^2	$2pq$	q^2

- Assume the allele frequency of A is the same in females and males.
- The probability of a baby frog to have A/A genotype is p^2
- What other assumptions are needed?
- Will any evolutionary force, such as migration and natural selection, affect the above expectation?

Hardy-Weinberg Law

- The prediction of genotype frequency from the allele frequency only relies on the step of forming new zygotes.
- For the prediction to hold, we just need:
    - random mating **with respect to the locus of interest**
    - same allele frequencies in both sexes
- Under these assumptions, the expected genotypic frequency distribution is achieved *instantaneously* upon mating
- Often hear about "no selection, no migration" only needed for HW frequencies to represent a **long term equilibrium**, known as Hardy Weinberg Equilibrium

---

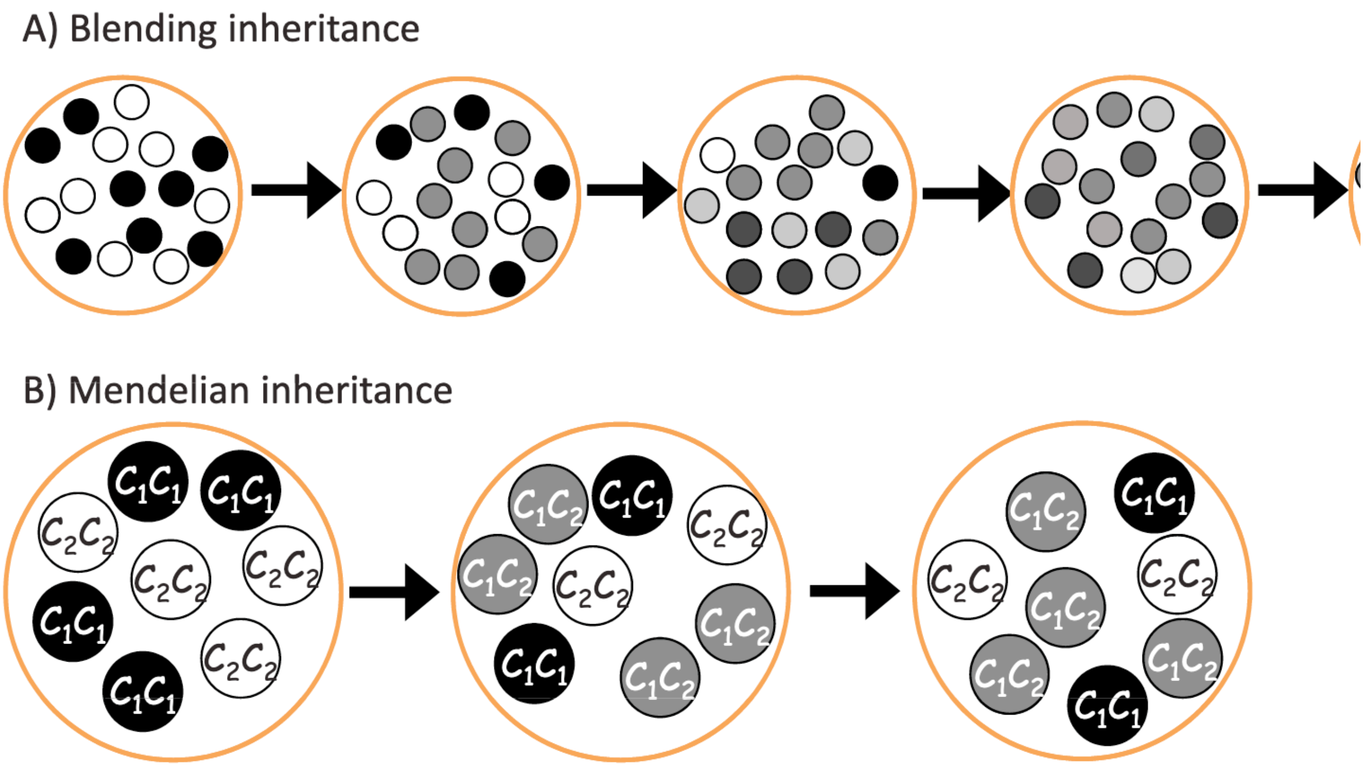
Consequences of Hardy Weinberg Law

1. Genotype frequencies can be estimated easily from allele frequencies.
2. Genetic variation can be maintained in a population.
3. There will be NO EVOLUTION when all of the following conditions are met
    - Infinite population size (no drift); no mutation; no selection; no migration; random mating; equal allele frequencies in males and females
    - when all of the above are true, allele frequencies do not change from generation to generation

---

History of Hardy-Weinberg Law

- Maybe the original Weinberg paper?



<%? footnotes %>

Using HW law to estimate heterozygote frequency

Example: cystic fibrosis, an autosomal recessive disease, has an incidence of 1 in 2500 in northern Europeans. Estimate the frequency of carriers.

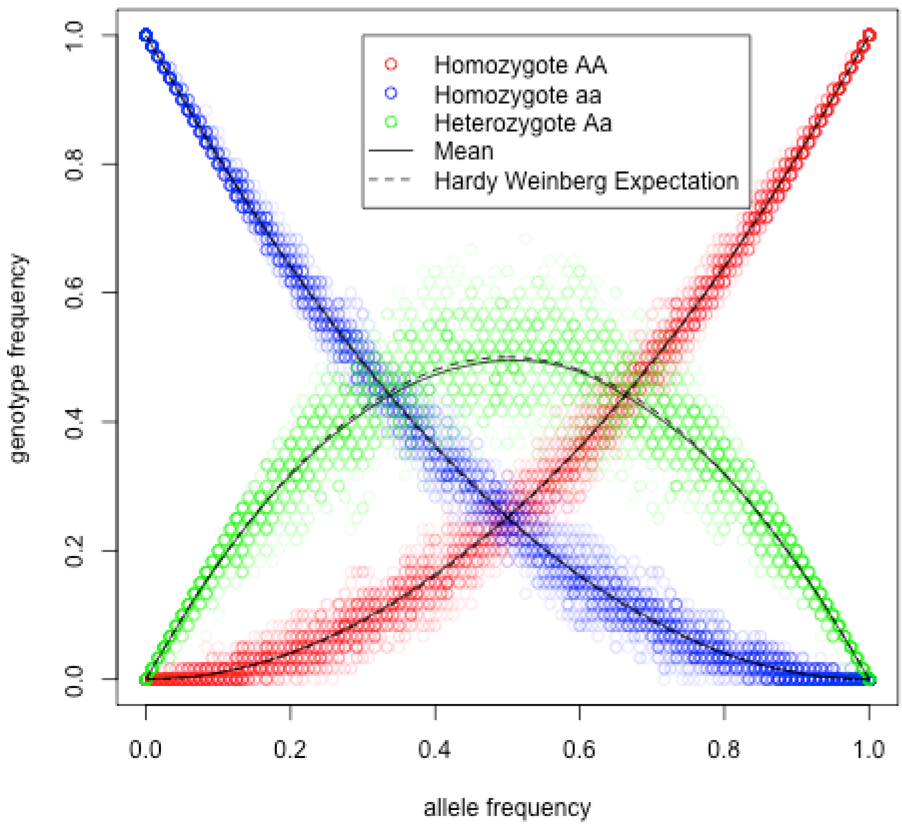
$q^2 = 1/2500$,

So, $q = 1/50 = 0.02, p = 1 - q = 0.98$

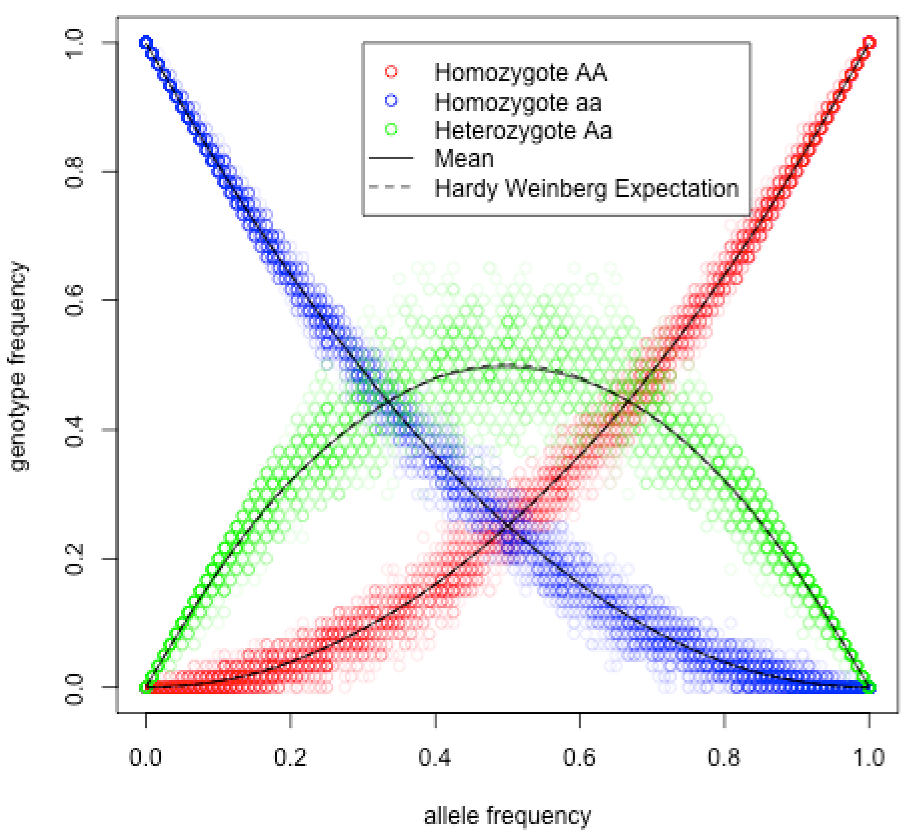
$2pq = 2 \times 0.98 \times 0.02 = 0.0392$, or 1 in 25 people.

Hardy-Weinberg Equilibrium

HapMap CEU (Europeans)



HapMap YRI (Africans)



- Solid line: mean genotype frequency; dashed line: HWE prediction. Source
- While HWE sounds too ideal to be a good fit to real data, researchers found that it actually holds remarkably well within populations. Since the main assumption for HWE to be true is that mating is random, how should one understand this?

HWE exercise

In a human population, the genotype frequencies at one locus are 0.5 AA, 0.4 Aa, and 0.1 aa. The frequency of the A allele is:

A) 0.20.

B) 0.32.

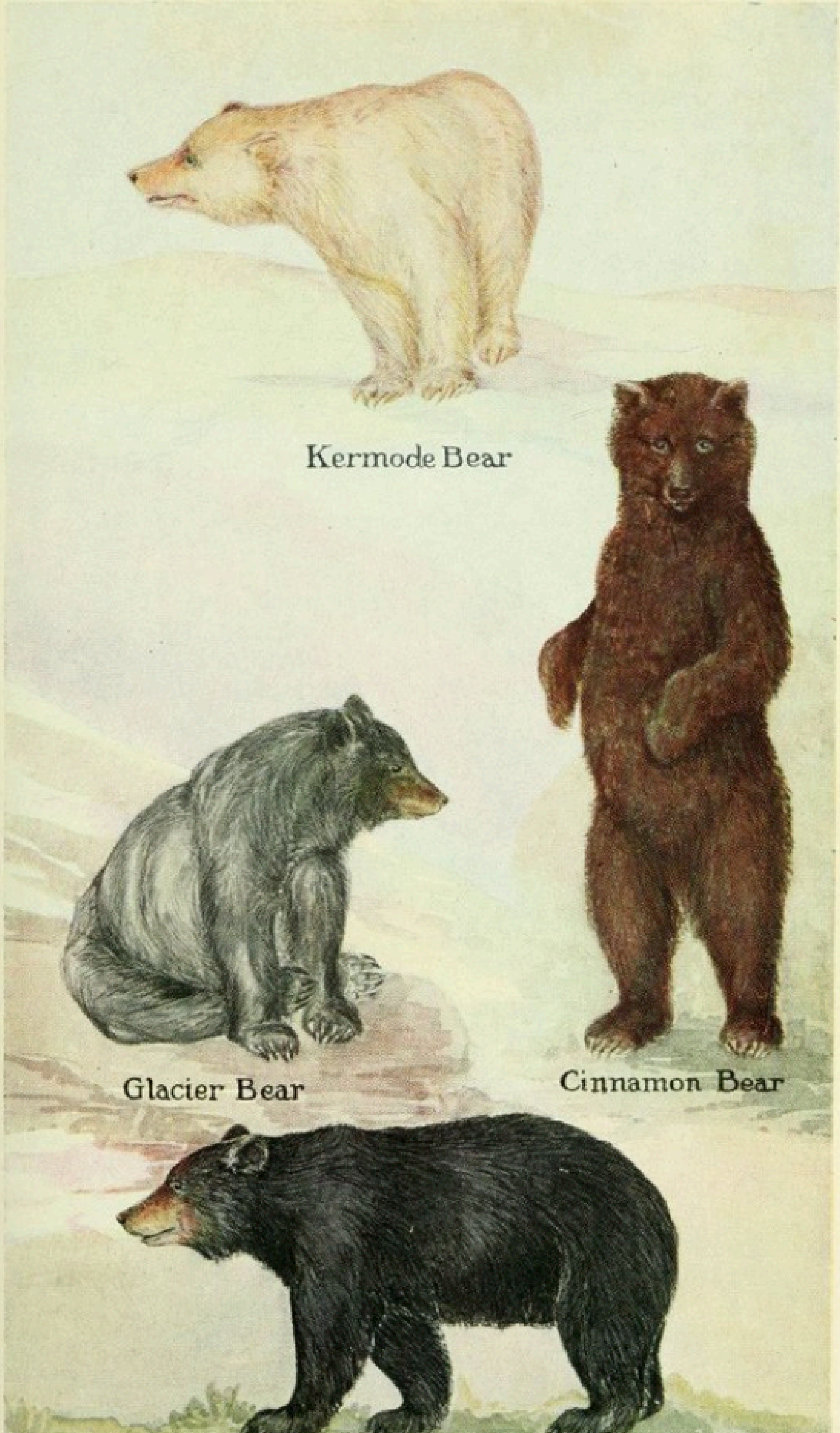
C) 0.50.

D) 0.70.

E) 0.90.

HWE exercise

PLATE IV



Kermode Bear

Glacier Bear

Cinnamon Bear

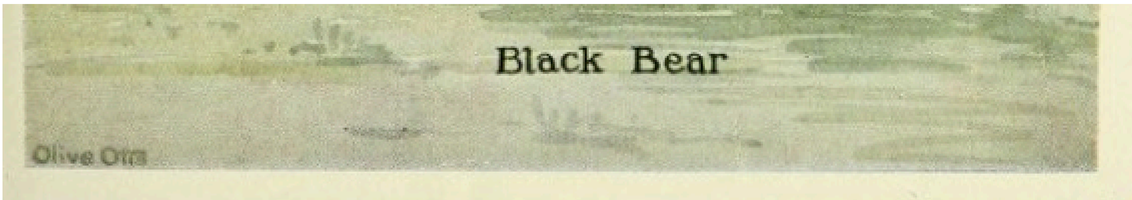


Figure 2.5: Kermode's bear (*Ursus americanus kermodei*). It's possible that this morph is favoured as the salmon these bears eat have a harder time seeing the light morph (Klinka and Reimchen, 2009). The adaptive value of tasting like cinnamon is unknown.

Hardy-Weinberg Law for X-linked loci

- How does HW Law apply to X-linked loci?

- Male-baldness is an X-linked trait.
- Androgen Receptor (AR) is an X-linked gene involved in male development.
- an AR haplotype called *Eur-H1* is **strongly associated with pattern baldness**.
- *Eur-H1* occurs at a frequency of 0.71 in Europe. What percentage of European men are expected to have this allele?
- European women?

Male pattern baldness

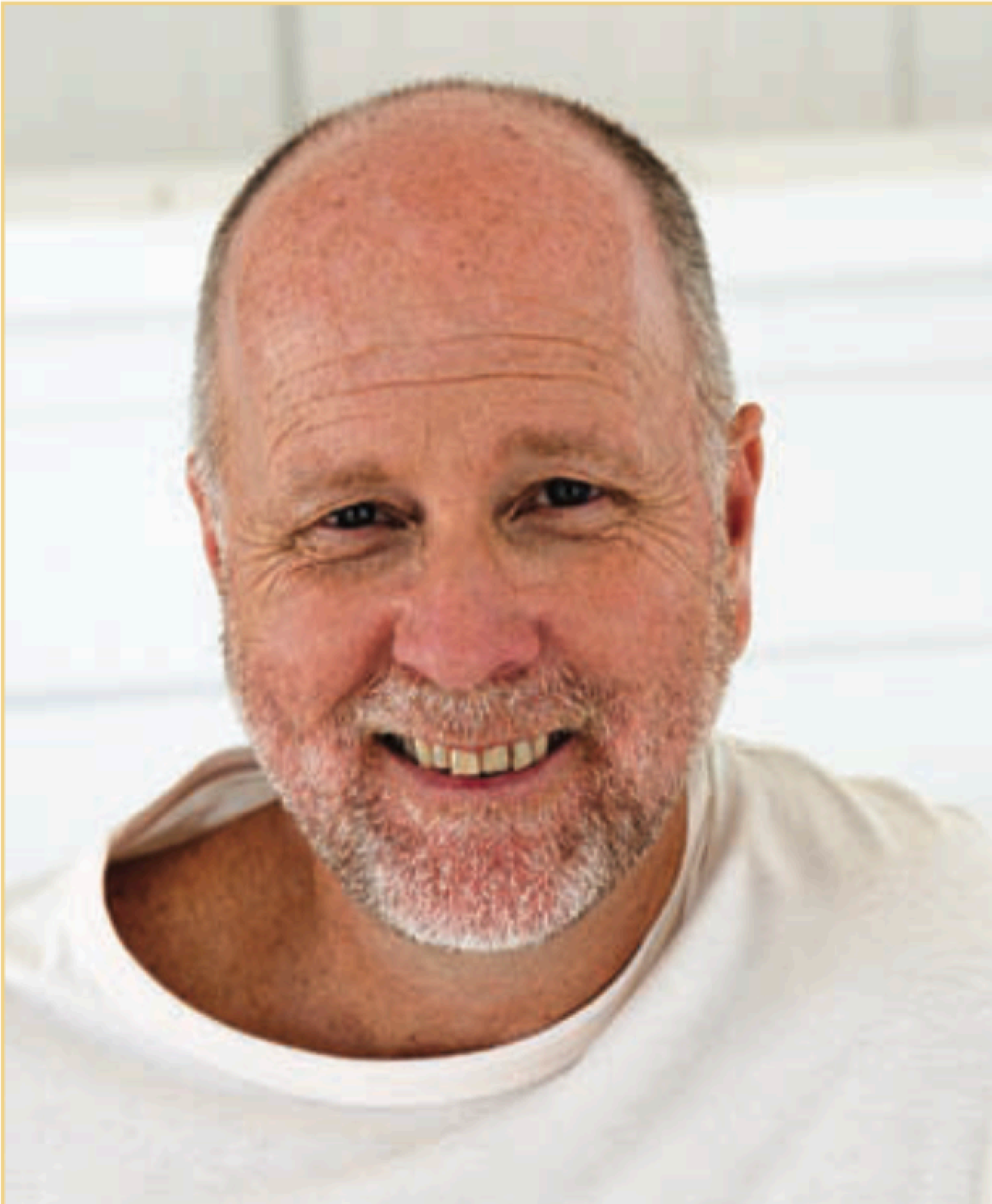


FIGURE 18-9 Individual showing male pattern baldness, an X-chromosome-linked condition. [*B2M Productions/Getty Images.*]

condition. [BLM Productions/Getty Images.]

-

Major violations to HWE

Assortative mating

- Non-random mating ***with respect to*** the genotype at a locus
- Ex: negative or disassortative mating
  - Major histocompatibility complex (MHC) locus, involved in immune response to pathogens, higher heterzygosity = more diversity ≈ higher fitness.
  - "sweaty shirt test": MHC alleles affect mating choice, prefer individuals with different MHC alleles

---

Major violations to HWE

isolation by distance

> Frequency variation for the FYnull allele of the Duffy blood group locus in Africa. [Data from P. C. Sabeti et al., Science 312, 2006, 1614–1620.]
>
> question
>
> what's the consequence of isolation by distance?

Major violations to HWE

inbreeding

- Mating between *close* relatives.
- How does inbreeding violate HWE?
- Why is inbreeding potentially harmful in outcrossing populations (like humans)?
- Selfing (extreme form of inbreeding) is common in animals and plants. What may be some advantages?
- Identical-by-Descent (IBD)

Measuring inbreeding

- **Inbreeding coefficient** (F): probability that two alleles in an individual trace back to the same copy in a common ancestor.
- **Identical-by-Descent** (IBD): define two alleles to be IBD if they are identical due to transmission from a common ancestor in the past few generations
- A "closed loop" indicates inbreeding.

Measuring inbreeding

- What's the probability that individual I is inbred? (calculating its inbreeding coefficient)

- The alleles transmitted through each mating are labeled x, t, w and z. Use "~" to indicate IBD.
- We wish to calculate $P(w \sim x)$
- $P(x \sim t) = \frac{1}{2}; P(w \sim z) = \frac{1}{2}$
- $P(z \sim t) = \frac{1}{2} + \frac{1}{2}F_A$
- $F_I = P(z \sim t) \times P(w \sim z) \times P(x \sim t) = (\frac{1}{2})^3(1 + F_A)$

Consequences of inbreeding

Examples of inbreeding effect on frequency of homozygous recessives

TABLE 18-3 Number of Homozygous Recessives per 10,000 Individuals for Different Allele Frequencies (q)

Mating	F	$q = 0.01$	$q = 0.005$	$q = 0.001$
Unrelated parents	0.0	1.00	0.25	0.01
Parent–offspring or brother–sister	1/4	25.75	12.69	2.51
Half-sib	1/8	13.38	6.47	1.26
First cousin	1/16	7.19	3.36	0.63
Second cousin	1/64	2.55	1.03	0.17

[1-1]

[1-2]: Introduction to Genetic Analysis, ed 11, Table 18-3

Consequence of inbreeding

> Frequency of genetic disorders among children of unrelated parents (blue columns) compared to that of children of parents who are first cousins (red columns).
>
> Data from C. Stern, Principles of Human Genetics, W. H. Freeman, 1973.

Review of Mon lecture

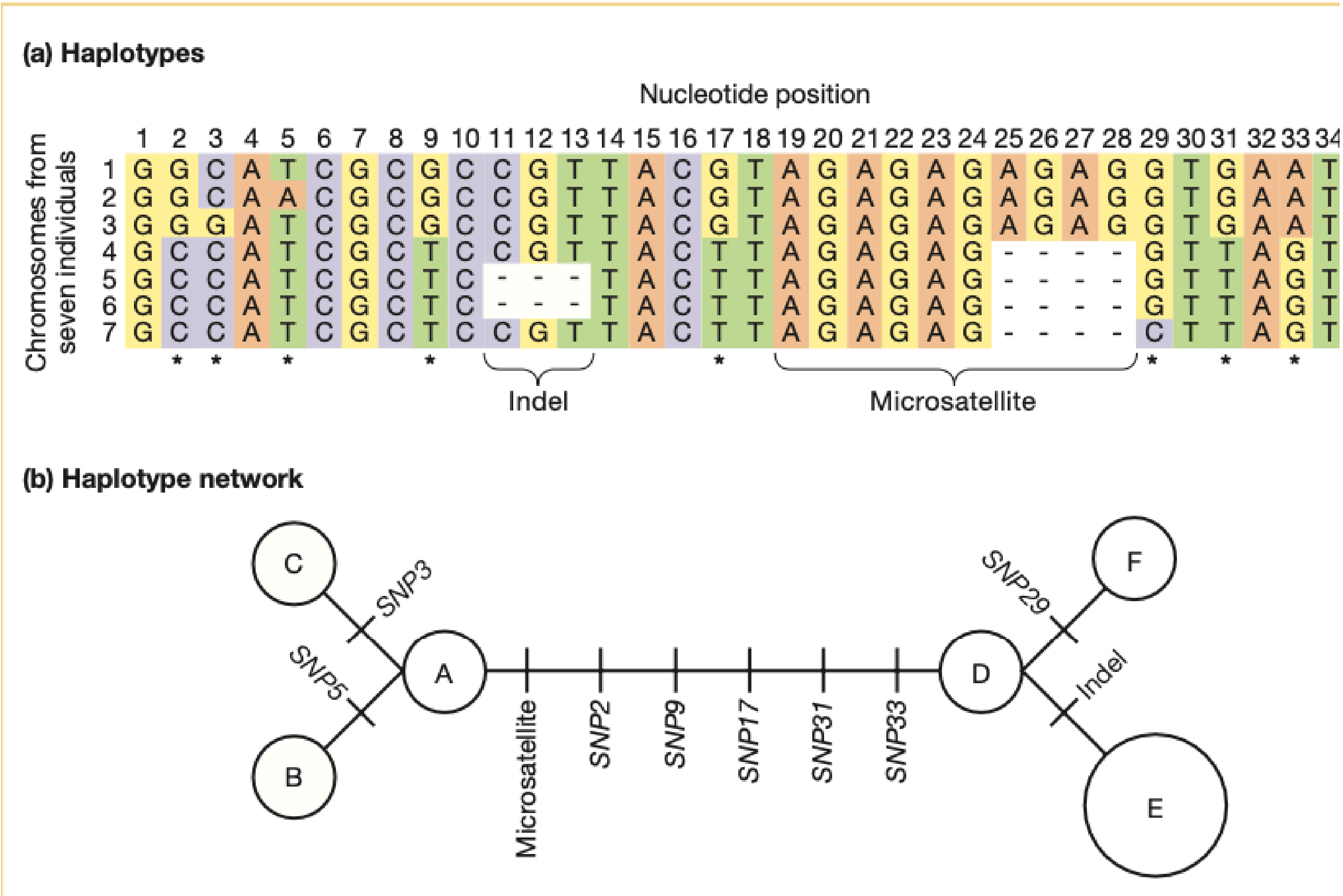
Question

1. Describe Hardy-Weinberg Equilibrium according to your understanding. What does it concern and what are its key assumptions?
2. Describe two ways for measuring the level of genetic variation.
3. Draw a cartoon haplotype network to describe the two hypotheses in the journal club paper (*to what extent populations within continental regions exist as discrete genetic clusters versus as a genetic continuum*)

note: $\pi = \sum_{i=1}^{n}(1 - \sum_{j=1}^{m}(p_{ij}^2))$

How do you quantify the level of genetic variation?

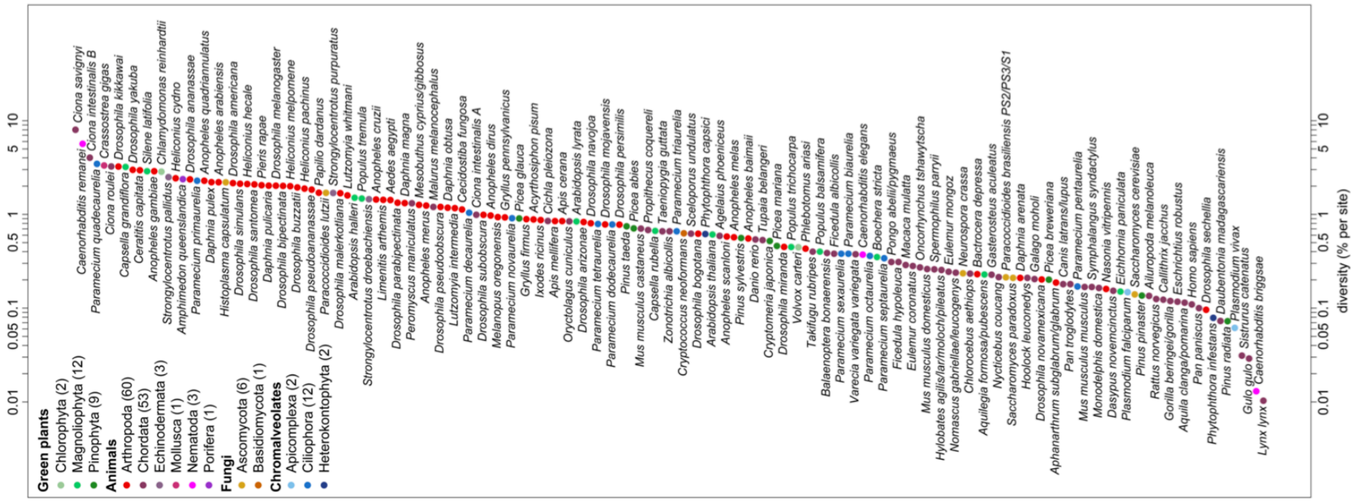
A haplotype network shows the relationship among haplotypes



- count the number of variable sites

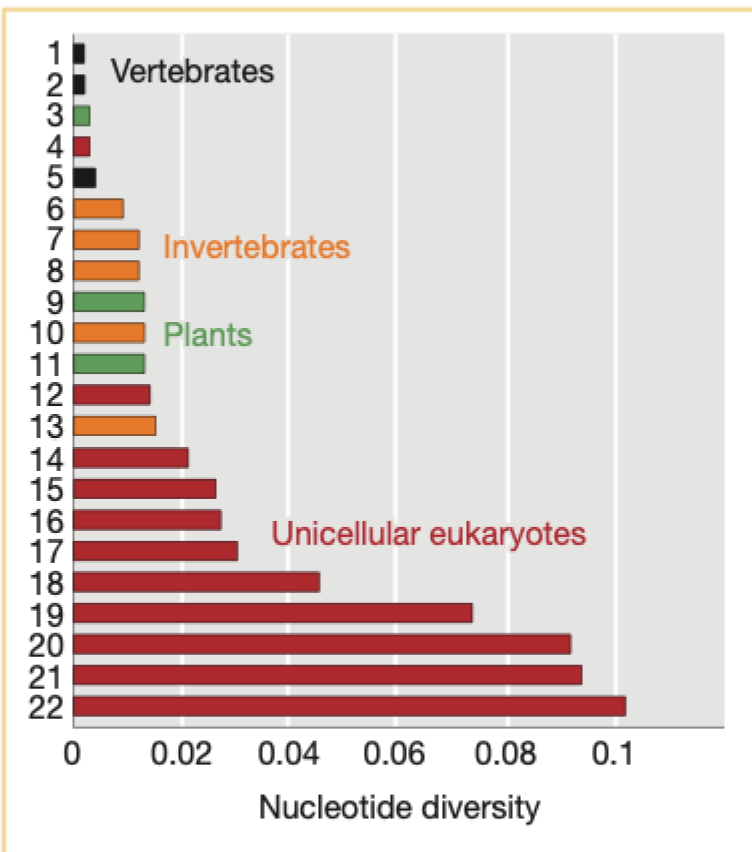
- account for the length of the region (sequence)
- account for the number of sequences (sample size)
- Average # of differences between two sequences

Level of genetic variation vary across species



[1-3]

[1-4]: Leffler et al. 2012. “Revisiting an Old Riddle: What Determines Genetic Diversity Levels within Species?” *PLoS Biology* 10 (9): e1001388. https://doi.org/10.1371/journal.pbio.1001388.



M. Lynch and J. S. Conery, Science 2003

What forces shape the level and pattern of genetic variation?

- How do new alleles enter the gene pool?
- What forces drive changes in the frequency of alleles?
- How can existing genetic variation recombine to create novel combinations of alleles?
- Forces examined next
  1. mutation
  2. migration
  3. drift
  4. recombination
  5. selection

Mutation

- **Mutation rate** (μ): probability that a copy of an allele changes to some other allelic form *in one generation*

- doesn't just apply to a single nucleotide position - for any locus
- in a single generation is important - other forces can change allele forms, but they act after the formation of zygote stage
- data in human (circa 2009) gave an estimate of $\sim 3.0 \times 10^{-8}$ mutations/nucleotide/generation for a part of the Y-chromosome. if we extrapolate this to the entire human genome, we get an estimate of about 100 new mutations one would inherit from each of our parents *on average*.

Migration

- In addition to mutation, migration is the other way by which new variation can be introduced into a population (not counting **new combinations**, see recombination)
- when there is isolation by distance, migration (or gene flow) is a homogenizing force, preventing allele frequencies from diverging too far

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Recombination

- Alleles are not gained or lost, but new combinations (haplotypes) are created by "mixing" existing ones.

- Let's consider the observed and expected frequencies of the four possible haplotypes for two loci, each with two alleles (A and B, with alleles A/a and B/b)
    - the four possible haplotypes are?
    - if the frequencies of the alleles are p_A and p_B, what are the expected frequencies for the four haplotypes naively?
    - $P_{AB} = p_A \times p_B$ etc.
    - If the observed frequencies are very close to the expected, we say that the two loci are in **linkage equilibrium**
    - If not, then the loci are said to be in **linkage disequilibrium (LD)**.

Linkage Disequilibrium

- Define LD as the difference (D) between the observed and expected frequencies
    - $D = P_{AB} - p_A p_B$
- Exercise: calculate D for the two scenarios on the left

Linkage Disequilibrium

- How LD arises
    - consider a population with only AB and Ab. what happens when a new allele "a" arises at A locus, on a chromosome that already possess b. Is LD zero? or is it maximum?
    - other than mutation, can migration cause LD? how?

- How LD decays
    - decays proportional to **time** and **recombination fraction**
    - can be used to estimate the age of mutation, how?

note:

- An allele in the G6PD gene called A^- leads to strongly reduced enzyme activity, and individuals who carry this allele develop hemolytic anemia. However, this allele also confers a 50 percent reduction in the risk of severe malaria in carriers.

Genetic drift and population size

- When calculating the expected genotype frequency from allele frequencies under Hardy-Weinberg Equilibrium, we made a sneaky assumption - the population size is assumed to be so large that we can "sample with replacement".

- Real populations have finite population sizes. Sampling can lead to fluctuations in the allele frequency from generation to generation
  - let's consider an extreme case: a population consisting of a single heterozygous (A/a) individual (N=1) at generation t_0. What's the allele frequency at t_0?
  - suppose this species can self-fertilize, and that the population size remains at one in the next generation (t_1). what's the probability that the allele frequency will change ("drift") to something other than its original value?
  - what happens when we increase N to 2?

note: when N=1, there is a 50% chance that one of the two alleles will be fixed in the next generation. when N=2, the chance is 12.5%

Genetic drift and population size

Drift
: any change in allele frequencies due to sampling error, not just loss or fixation of an allele

> **Question**
>
> In a population with 500 individuals (N=500) and two alleles at a frequency of $p = q = 0.5$, if the next generation has 501 copies of A and 499 copies of a, **has there been genetic drift**?

Genetic drift and population size

- Drift is caused by sampling variation In a **finite population**.

- When drift is operating, one can **calculate the probabilities of different outcomes**, but one **cannot accurately predict the outcome that will occur**
- At a locus, drift continues to affect the allele frequency from generation to generation, until one allele is eventually fixed (and all others are lost), hence no more variation at that locus.
- Drift **doesn't proceed in a specific direction**.

Genetic drift and population size

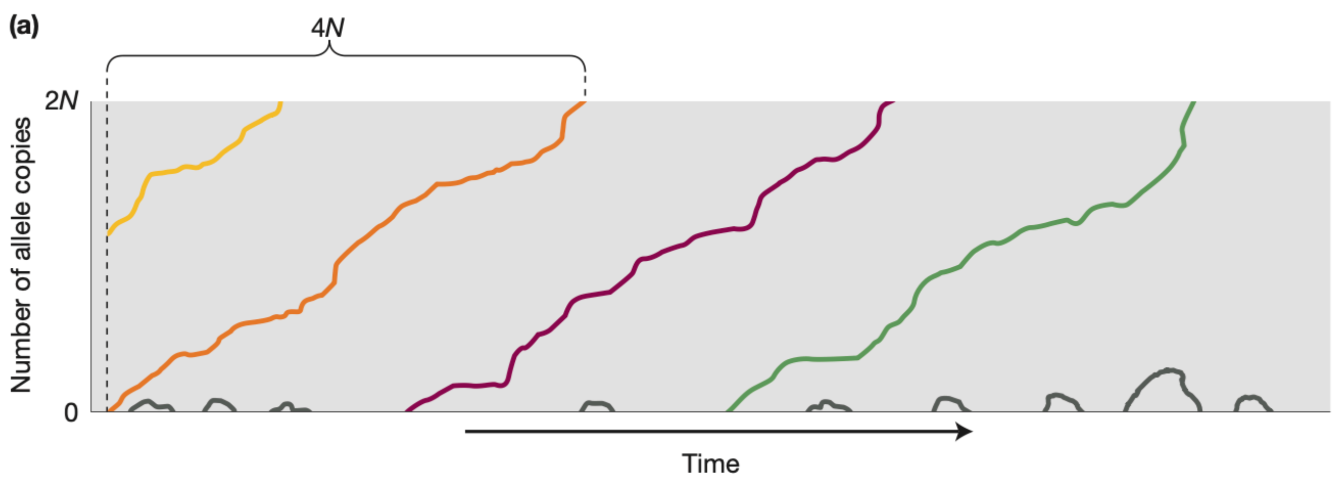
![[20240509-IntroGenetAnal-18-18a.png]] ![[20240509-IntroGenetAnal-18-18b.png]] ![[20240509-IntroGenetAnal-18-18c.png]]

note:

1. randomness of the processes - no two trajectories are identical
2. when N=10, a lot more populations became fixed for one of the two alleles (5/6), compared with 0/6 when N=500, shows that the effect of drift is inversely proportional to the size of the population

Genetic drift and population size

- Prob(an allele goes to fixation) ~ frequency in the present generation
- The initial frequency of a new allele in a diploid population of size N is $\frac{1}{2N}$
- If N is even modestly large, e.g., 10^4, the probability of its fixation is 5×10^{-5}, while the probability of it eventually being lost is close to 1.0!



note: most new mutations are lost; average time to fixation | fixed is 4N

Consequence of genetic drift

Slightly deleterious alleles may be driven to fixation, while beneficial mutations are subject to loss

1. Strongly deleterious mutations are quickly lost. But slightly deleterious ones can drift to fixation - and when it does, it does so **quickly** (counterintuitive?)
   - do you know that the fate of a mutation is not only determined by its functional impact (beneficial or deleterious), but also population size?
2. When a new beneficial mutation arise, there is an appreciable chance that it will be lost in the first few generations!
   - the individual carrying the new mutation may not have the opportunity to reproduce.
   - or it may not pass on the beneficial mutation (since by definition, a new mutation must be a single copy in a diploid individual)

Molecular clock: implication of genetic drift

- Because of genetic drift, neutral mutations can become fixed, which we call "substitutions".

- note the difference between "mutation" and "substitution"
- the number of mutations that are expected to "appear" in a population in one generation is proportional to the number of copies in the gene pool, i.e., $2N\mu$, where μ denotes the mutation rate per locus per generation.
- recall that the probability of fixation for a neutral mutation equals its initial frequency. for a new, neutral mutation, this is $\frac{1}{2N}$
- therefore, the number of **fixed neutral mutations** per generation, which we will call "substitution rate", or k, is $k = 2N\mu \times \frac{1}{2N} = \mu$
- that is, the substitution rate for neutral mutations equal the rate at which they appear. this is only true for neutral mutations!

Molecular clock: implication of genetic drift

- d: number of **neutral substitutions** in a gene in both species since their divergence.

- $d = 2tk$, where t is the divergence time.
- $t = \frac{d}{2k}$, this provides a way to estimate divergence time based on the number of substitutions and estimate of substitution rate.
- the estimated divergence time is in the unit of generations.

- how to determine which substitutions are neutral?
- can we observe all neutral substitutions?
- how do we know mutation rate?

Genetic drift and population size

Key concept

- Population size is a key factor affecting genetic variation in populations.
- Genetic drift is a stronger force in small populations than in large ones.
- The probability that an allele will become fixed in (or lost from) a population by drift is a function of its frequency in the population and population size.
- Most new mutations are lost from populations by drift.
- The constant rate of substitution for neutral mutations can be used to estimate divergence time.

Selection

Natural selection (*Darwinian*)

Populations change (*evolve*) over time as the environment (*nature*) favors (*selects*) features that enhance the ability to survive and reproduce.

Selection

Question

We often use the phrase "*survival of the fittest*" to describe Darwinian selection. If an individual is physically strong, resistant to disease, and lives a long life, does this individual have a higher fitness?

Selection

- **Natural selection** means individuals with certain features are more likely to *survive AND reproduce* than other individuals lacking these features.
- Mutation generates new variants; recombination creates new combinations (haplotypes), both of which affect the phenotype (feature).
- Individuals with higher **fitness** are more likely to survive and reproduce, i.e., their alleles will have a higher frequency in the gene pool in next round of reproduction -> changing the allele frequency.
- Where does selection operate in the left diagram?

How natural selection alters allele frequency?

> Figure 10.1: The rapid evolution of drug-resistant SHIV. The viral load of SHIV in the blood of a macaque (black line), the frequency of a drug resistance mutation (red line). Data from Feder et al. (2017).

How natural selection alters allele frequency?

- Consider a locus A with three genotypes, A/A, A/a, and a/a in a population. The initial allele frequencies are $p = 0.1, q = 0.9$
- A is a favored dominant allele
Genotype	A/A	A/a	a/a
Average number of offspring (W)	10	10	5
Relative fitness (w)	1.0	1.0	0.5
Genotype frequency			

How natural selection alters allele frequency?

- A is a favored dominant allele
	A/A	A/a	a/a

Average number of offspring (W)	10	10	5
Relative fitness (w)	1.0	1.0	0.5
Genotype frequency	0.01	0.18	0.81

- The relative contribution of each genotype to the gene pool is determined by **the product of its fitness and frequency**

Genotype	A/A	A/a	a/a	Sum
Relative contribution	1 x 0.01	1 x 0.18	0.5 x 0.81	
	= 0.01	= 0.18	= 0.405	0.595

How natural selection alters allele frequency?

- The relative contribution of each genotype to the gene pool is determined by **the product of its fitness and frequency**

Genotype	A/A	A/a	a/a	Sum
Relative contribution	0.01	0.18	0.405	0.595
Genotype frequency	0.02	0.30	0.68	1.0

- Now, let's use the genotype frequency to calculate the allele frequency, and then use that to calculate the expected genotype frequency in the next generation of zygotes under HWE
  - $p' =?$ how much did p change over one round of selection?
  - $P_{AA} =?$, etc.

note:

1. $p' = 0.02 + 0.15 = 0.17$
2. $P_{AA} = 0.17^2 = 0.0289, P_{Aa} = 2 \times 0.17 \times 0.83 = 0.2822, P_{aa} = 0.83^2 = 0.6889$

How natural selection alters allele frequency?

More generally

define $\bar{w} = p^2 w_{A/A} + 2pq w_{A/a} + q^2 w_{a/a}$ is the mean relative fitness of the population

then, $p' = \frac{p^2 w_{A/A} + \frac{1}{2} \times 2pq w_{A/a}}{\bar{w}} = p \frac{p w_{A/A} + q w_{A/a}}{\bar{w}}$

we define $w_A=pw_{A/A}+qw_{A/a}$ as allelic fitness, or mean fitness of allele A

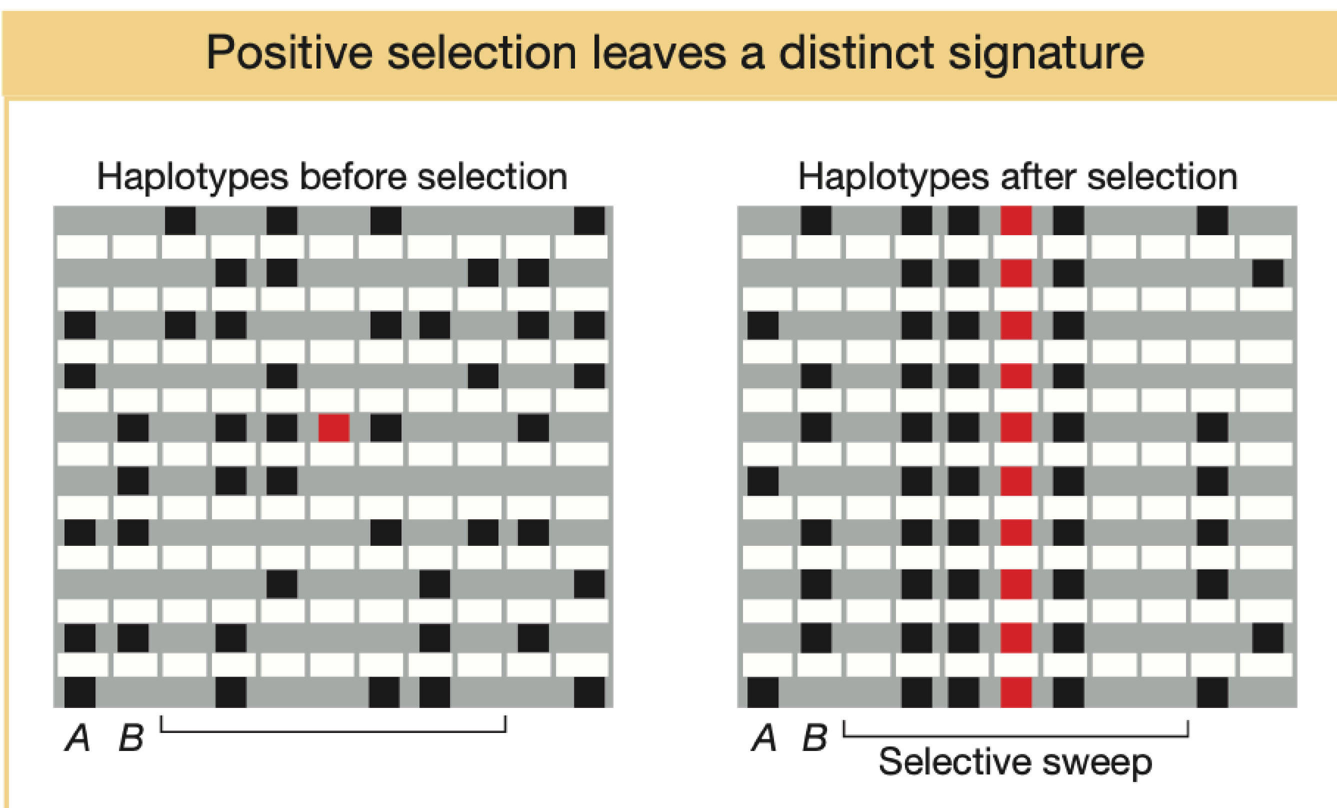
$\Delta p=p'-p=p\frac{w_A}{\bar{w}}-p=\frac{p(w_A-\bar{w})}{\bar{w}}$

Forms of selection

1. Directional selection
    - Positive selection
    - Purifying (negative) selection

2. Balancing selection
    - if the heterozygous class has higher fitness than either homozygous ones, selection would favor the **maintenance of both alleles**.
    - can you think of example scenarios for balancing selection?
    - are there other types of selection that can **maintain genetic diversity** rather than **eliminating it**?

Signatures of positive selection



[1-5]: Introduction to Genetic Analysis, ed 11, Fig. 18-22

Signatures of positive selection

- By driving the favored allele to fixation "very fast", positive selection causes a selective sweep that ends up
  - also fix linked sites or drive the linked allele to high frequency
  - hence removing genetic variation from the population
  - creates LD
- Note that this signature is **transient**. What does this mean, why?

Use selective sweep signature to infer positive selection

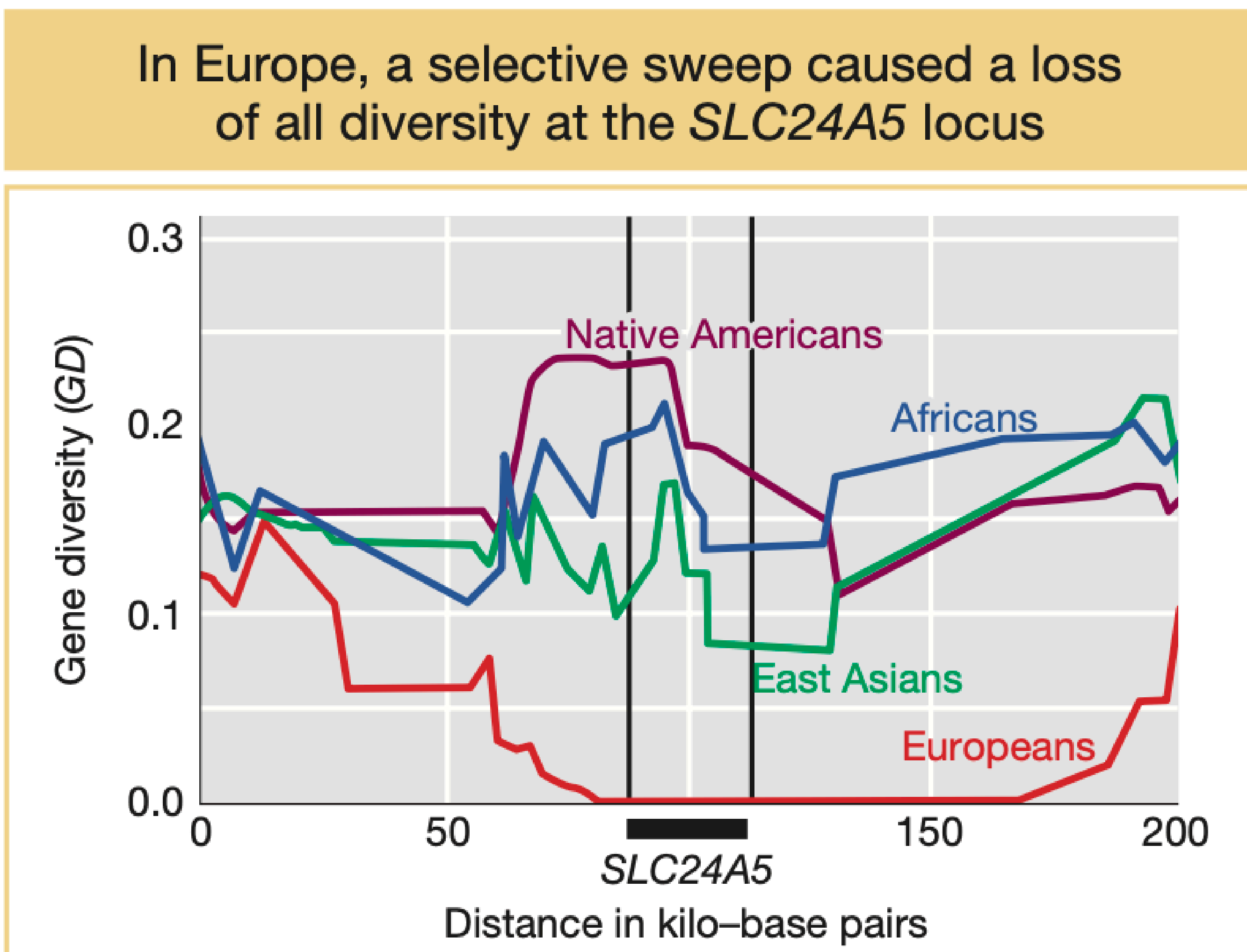
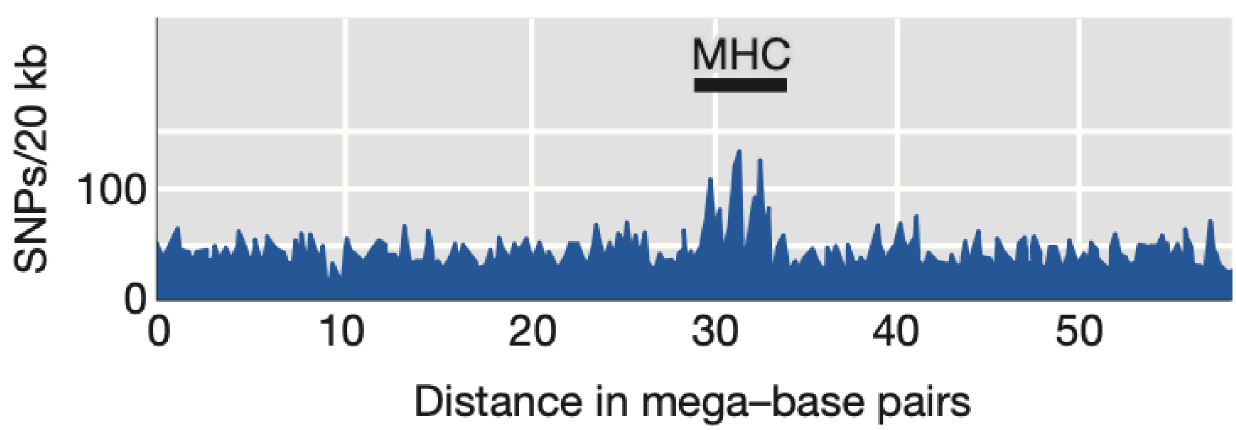


TABLE 8-6 Some Genes Showing Evidence for Natural Selection in Specific Human Populations

Gene	Presumed Trait	Population
EDA2R (ectodysplasin A2 receptor)	Male pattern baldness	Europeans
EDAR (ectodysplasin A receptor)	Hair morphology	East Asians
FY^{null} (Duffy antigen)	Resistance to malaria	Africans
G6PD (glucose-6-phosphate dehydrogenase)	Resistance to malaria	Africans
Hb (hemoglobin B)	Resistance to malaria	Africans
KITLG (KIT ligand)	Skin pigmentation	East Asians and Europeans
LARGE (glycosyltransferase)	Resistance to Lassa fever	Africans
LCT (lactase)	Lactase persistence; ability to digest milk sugar as an adult	Africans, Europeans
LPR (leptin receptor)	Processing of dietary fats	East Asians
MC1R (melanocortin receptor)	Hair and skin pigmentation	East Asians
MHC (major histocompatibility complex)	Infectious disease resistance	Multiple populations
OCA2 (oculocutaneous albinism)	Skin pigmentation and eye color	Europeans
PPARD (peroxisome proliferator-activated receptor delta)	Processing of dietary fats	Europeans
SI (sucrase-isomaltase)	Sucrose metabolism	East Asians
SLC24A5 (solute carrier family 24)	Skin pigmentation	Europeans and West Asians
TYRP1 (tyrosinase-related protein 1)	Skin pigmentation	Europeans

Source: P. C. Sabeti et al., *Science* 312, 2006, 1614–1620; P. C. Sabeti et al., *Nature* 449, 2007, 913–919; B. F. Voight et al., *PLoS Biology* 4, 2006, 446–458; J. K. Pickrell et al., *Genome Research* 19, 2009, 826–837.

Balancing selection can lead to regions of unusually high genetic diversity



1. Introduction to Genetic Analysis, ed 11, Fig. 18-24 ↩ ↩ ↩ ↩ ↩ ↩